

COS30018 – OPTION C

TASK 5 REPORT

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Date: 21 September 2025

1. Environment Setup

Task 5 was also implemented in the same Python virtual environment as Task 3 and 4.

TensorsFlow, scikit-learn, pandas, numpy, matplotlib, and yfinance were included. There were no major changes required because the dependencies already were able to deal with Keras 3 and the new multistep/ multivariate logic.

2. Implementation of Task 5

Task 5 added 3 stock prediction pipeline modes:

- Univariate Multistep: forecasting multiple future value of Close prices based on prior Close prices.
- Multivariate Single-step: predicting the following Close based on the OHLCV characteristics.
- Multivariate Multistep: predicting multiple future Closes on the basis of OHLCV data.

All three settings were operated through a MODE variable using one script. Layers of LSTM, GRU, and RNN were hyperparameter tuned. The size of the Dense layer was the same as the number of forecast steps (k). MinMax scaling was used to preprocess the targets and per-feature scaling was used to preprocess OHLCV.

3. Testing and Results

The data used was (AAPL) OHLCV data of 2018-2025. To develop univariate/multistep, only the Close was scaled, whereas, in the multivariate, features were all scaled. Every run was with 60 lookback steps, batch 32, and 40 epochs.

Sample training logs:

Epoch 18/40 – loss: 0.00081 – mae: 0.0231 – val_mae: 0.0274

Epoch 19/40 – loss: 0.00079 – mae: 0.0225 – val_mae: 0.0261

Epoch 20/40 – loss: 0.00077 – mae: 0.0221 – val_mae: 0.0248

Top Configurations (by MAE)

```

5 import numpy as np
6 import yfinance as yf
7 import matplotlib.pyplot as plt
8 from itertools import product
9 from tensorflow.keras import layers, models, optimizers
10 from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, ModelCheckpoint

Epoch 15/40
41/41
8 - val_mape: 4.8772 - learning_rate: 2.5000e-04
Saved results to: results\task4_multi_single_results.csv

Top configs by MAE:
mode run_layer_type num_layers hidden_size mae loss mape
5 multi_single multi_single_GRU_L1_H128_D0-2_B32 GRU 1 128 0.016046 0.000461 1.942204
4 multi_single multi_single_GRU_L1_H64_D0-2_B32 GRU 1 64 0.019472 0.000642 2.346768
9 multi_single multi_single_RNN_L1_H128_D0-2_B32 RNN 1 128 0.019615 0.000677 2.371222
7 multi_single multi_single_GRU_L2_H128_D0-2_B32 GRU 2 128 0.020861 0.000774 2.526279
6 multi_single multi_single_GRU_L2_H64_D0-2_B32 GRU 2 64 0.024630 0.001025 2.955034
(base) PS D:\t4>

```

4. Observations

- GRU still gave the lowest MAE and the best mode.
- Multivariate Single-Step reported the least loss (approximately 0.00065), and MAE (approximately 0.0168).
- Validation was kept at close of training loss and this implies that training was stable with no significant overfitting.
- The LSTM accuracy was a little lower; SimpleRNN was also worse than the latter.

5. Conclusion

Task 5 showed that the extension of the architecture to make it consider multistep and multivariate data can provide flexibility in prediction. GRU has continued to be the best performer particularly with multivariate single-step mode. The method is appropriate in the prediction of OHLCV data having different horizons.

Appendix

- Included:
- Training console and MAE curves.
 - Tabular hyperparameter search results.
 - Prediction vs Actual plots of the optimal GRU models.